



**Università
di Genova**

DIBRIS DIPARTIMENTO
DI INFORMATICA, BIOINGEGNERIA,
ROBOTICA E INGEGNERIA DEI SISTEMI

High Performance Ecoinformatics

Enrico Pezzano

Master Thesis

Università di Genova, DIBRIS Via Opera Pia, 13 16145 Genova, Italy
<https://www.dibris.unige.it/>



MSc Computer Science
Data Science and Engineering Curriculum

High Performance Econinformatics

Enrico Pezzano

Advisor: Prof. Daniele D'Agostino
Prof. Gabriele Casazza (DISTAV)

Examiner: Prof. Matteo Dell'Amico

September, 2026

DEDICATION

Abstract Ecoinformatics is a modern method of doing science that enables scientists to generate new knowledge through innovative tools and approaches. Ecology is increasingly becoming a data-intensive science, involving the acquisition, preservation, and analysis of vast amounts of data from a variety of sources, including satellite and airborne remote sensing, instruments, sensors, and human observations. Ecologists are increasingly addressing larger-scale questions of both scientific and societal relevance, such as climate change and its consequences for ecosystems and related ecosystem services in the future. The topic of this thesis project stems from a research study conducted by the Department of Earth, Environmental, and Life Sciences (DISTAV, University of Genoa) that analyzes the shift in the distribution of natural habitats in the Alps due to future climate change to support decision-making processes regarding conservation planning and prioritization. The analysis is conducted using Species Distribution Models (SDMs), a dataset of approximately 3 million data points, and future climate change scenarios based on General Circulation Models (climate models). The statistical software R (<https://www.r-project.org/>) is used to generate future predictions of habitat distribution changes and biodiversity loss. The thesis candidate will work on improving the script and algorithmic functions, written in R, to streamline model analysis based on their knowledge and computer science background. This thesis project leverages interdisciplinary collaboration between DIBRIS and DISTAV to achieve better scientific results. Depending on the candidate's interest and quality of results, the thesis can be upgraded with asterisk.

Table of Contents

Chapter 1	Introduction and Motivation	10
1.1	Project context (taken from <code>Methods_GC_LD.pdf</code>)	10
Chapter 2	Background	12
2.1	Ecoinformatics	12
2.2	Biological study context	12
2.3	Data used	13
2.4	Related work	13
Chapter 3	Code as is	14
3.1	R	14
3.2	BIOMOD2	14
3.3	Code pipeline	14
3.4	Original execution times	14
3.5	Improved execution times	14
3.6	Preliminary measurements on Spartaco	14
Chapter 4	My new code	16
4.1	Container	16
4.2	Parallel model execution	16
4.3	Parallel I/O	16

Chapter 5	Experimental results	17
5.1	Leonardo	17
5.2	Tests	17
5.3	Discussion and lessons learned	17
Chapter 6	Conclusions and future development	18
6.1	Conclusions	18
6.2	Future development	18
6.3	Ecoinformatics	18
6.4	Data used	18
6.5	Computer science objective	18
6.6	Interdisciplinary collaboration	19
Chapter 7	Appendix	20
	Bibliography	21

Chapter 1

Introduction and Motivation

introduzione deve essere scritta alla fine

1.1 Project context (taken from **Methods_GC_LD.pdf**)

Proposta tesi ecoinformatics Lucia.

L'ecoinformatics è un metodo moderno di fare scienza che consente agli scienziati di generare nuove conoscenze attraverso strumenti e approcci innovativi. L'ecologia sta diventando in maniera crescente una scienza ad alta intensità di dati che comporta l'acquisizione, la conservazione e l'analisi di enormi quantità degli stessi provenienti da una serie di fonti, tra cui il telerilevamento satellitare e aereo, strumenti, sensori e osservazioni umane. Gli ecologisti affrontano sempre più spesso questioni su scala più ampia che hanno rilevanza sia scientifica che sociale come, ad esempio, i cambiamenti climatici e le conseguenze che tali avranno sugli ecosistemi, e servizi ecosistemici a loro connessi, nel futuro. Il tema di questo progetto di tesi ha origine da uno studio di ricerca portato avanti dal Dipartimento di Scienze della Terra, dell'Ambiente e della Vita (DISTAV, Università degli Studi di Genova) che analizza il cambio nella distribuzione di habitat naturali sulle Alpi a causa di cambiamenti climatici futuri per supportare processi decisionali riguardo la pianificazione e prioritizzazione per la conservazione. L'analisi viene fatta utilizzando gli Species Distribution Models (modelli di distribuzione spaziale delle specie), un dataset di circa 4 milioni di dati, e scenari futuri di cambiamento climatico basato sul calcolo dei General Circulation Models (modelli climatici). Il software statistico R (<https://www.r-project.org/>) viene utilizzato per generare predizioni future del cambio di distribuzione di habitat e della perdita di biodiversità. Il tesista si occuperà di apportare miglioramenti allo script di lavoro, alle funzioni algoritmiche, per rendere efficiente l'analisi dei modelli in base alle

proprie conoscenze e studi informatici. Questo progetto di tesi vede una collaborazione interdisciplinare tra il DIBRIS e il DISTAV come punto di forza per ottenere risultati migliori nel campo scientifico.

Chapter 2

Background

real world and eventually related work

2.1 Ecoinformatics

2.2 Biological study context

An internal DISTAV methods document describes the biological study supported by the computational workflow. The study examines how climate change may alter the suitability of Alpine grassland habitats. It considers ten dry, alpine and subalpine habitat types from the EUNIS classification and 167 diagnostic plant species. Occurrence records are associated with environmental predictors at a spatial resolution of approximately 1 km. The predictors describe climate, soil and topographic heterogeneity. Species Distribution Models combine five modelling techniques and ensemble forecasts to estimate present and future habitat suitability.

The biological protocol also defines the analyses that follow model projection: changes in suitability, spatial shifts, habitat-level hotspot and cold-spot maps, and the relationship between suitability changes and species tolerance. This document provides the scientific context for the computing work, but it is not treated as the exact specification of every run. Dataset sizes, model repetitions and available climate scenarios changed during the project; the computational chapters therefore report the configuration recorded by the corresponding scripts and run manifests.

2.3 Data used

2.4 Related work

Chapter 3

Code as is

Existing workflow and computational problem.

3.1 R

3.2 BIOMOD2

3.3 Code pipeline

3.4 Original execution times

3.5 Improved execution times

3.6 Preliminary measurements on Spartaco

Before the Leonardo experiments, Lucia manually recorded four trial runs on Spartaco. The worksheet records per-phase time and RAM observations for two species, with different pseudo-absence settings and CPU counts. Table 3.1 preserves selected values from those records.

The measurements are useful for reconstructing the starting point of the project. They are not a controlled benchmark: the worksheet does not identify the code revision, container,

hardware configuration, input version or measurement method. The source label `n_cells` is retained in the table because its meaning cannot be established from the worksheet alone. These results are therefore not compared directly with the later Leonardo measurements.

Table 3.1: Selected fields from four manually recorded trial runs on Spartaco. `n_cells` retains the source label. Time units are those recorded in the original worksheet; the maximum RAM is the largest phase value in that row.

Species	<code>n_cells</code>	PA	Rep.	RUN	CPU	Formatting	Models	Future projection	Max RAM (GB)
<i>Omalotheca hoppeana</i>	681	10000	10	10	5	1.9 h	11.33 min	3.65 min	52.9
<i>Omalotheca hoppeana</i>	681	10000	10	10	10	2.02 h	7.74 min	8.07 min	56.16
<i>Agrostis capillaris</i>	100000	5000	5	5	5	32.25 min	53.87 min	3.73 min	26
<i>Agrostis capillaris</i>	100000	10000	5	5	5	59.37 min	1.6 h	4.22 min	29.96

Chapter 4

My new code

Design and implementation of the improvements.

4.1 Container

4.2 Parallel model execution

4.3 Parallel I/O

Chapter 5

Experimental results

Experimental methodology and results.

5.1 Leonardo

5.2 Tests

5.3 Discussion and lessons learned

Chapter 6

Conclusions and future development

6.1 Conclusions

6.2 Future development

6.3 Ecoinformatics

What it is.

Why large datasets are used.

Why high-performance computers are used.

Purpose.

6.4 Data used

In the **Eval...** matrix, results with **calibration** below 0.5 are considered less performant.

6.5 Computer science objective

To be defined.

6.6 Interdisciplinary collaboration

Optional section.

Chapter 7

Appendix

Temporary or optional chapter.

Bibliography

- [CHE] CHELSA. CHELSA climatologies. Dataset documentation; accessed 2026-09-01. URL: <https://www.chelsa-climate.org/datasets/chelsa-climatologies>.
- [CINa] CINECA. Access to the systems. HPC documentation; accessed 2026-09-01. URL: <https://docs.hpc.cineca.it/general/access.html>.
- [CINb] CINECA. Leonardo. HPC documentation; accessed 2026-09-01. URL: <https://docs.hpc.cineca.it/hpc/leonardo.html>.
- [CINc] CINECA. Singularity and apptainer containers. HPC documentation; accessed 2026-09-01. URL: <https://docs.hpc.cineca.it/services/singularity.html>.
- [GBLPT26] Maya Guéguen, Hélène Blancheteau, Rémi Lemaire-Patin, and Wilfried Thuiller. *biomod2: Ensemble Platform for Species Distribution Modeling*, 2026. R package version 4.3-4-5. URL: <https://CRAN.R-project.org/package=biomod2>, doi:10.32614/CRAN.package.biomod2.
- [HBB26] Robert J. Hijmans, Andrew Brown, and Márcia Barbosa. *terra: Spatial Data Analysis*, 2026. R package version 1.9-11. URL: <https://rspatial.org/>.
- [KCB⁺17] Dirk N. Karger, Olaf Conrad, Jörg Böhner, Tobias Kawohl, Holger Kreft, Ruben W. Soria-Auza, Niklaus E. Zimmermann, H. Peter Linder, and Michael Kessler. Climatologies at high resolution for the earth’s land surface areas. *Scientific Data*, 4:170122, 2017. doi:10.1038/sdata.2017.122.
- [R C26] R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria, 2026. URL: <https://www.R-project.org/>.
- [TCA24] Matteo Turisini, Mirko Cestari, and Giorgio Amati. LEONARDO: A pan-european pre-exascale supercomputer for HPC and AI applications. *Jour-*

nal of Large-Scale Research Facilities, 8:A186, 2024. doi:10.17815/
jlsrf-8-186.